Repurchase Distribution Tilt and the Cross Section of Stock Returns

I. M. Harking

October 21, 2025

Abstract

This paper studies the asset pricing implications of Repurchase Distribution Tilt (RDT), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on RDT achieves an annualized gross (net) Sharpe ratio of 0.46 (0.41), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 45 (35) bps/month with a t-statistic of 4.76 (3.80), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Volume Variance, Price delay r square, Days with zero trades, Volume to market equity, Days with zero trades, Days with zero trades) is 29 bps/month with a t-statistic of 3.10.

1 Introduction

Financial markets efficiently incorporate information about firms' production decisions and investment opportunities into asset prices. The cross-section of stock returns reflects heterogeneous exposure to systematic risks arising from firms' real investment frictions and productivity shocks. Despite extensive research documenting how corporate payout policies relate to expected returns, we lack a comprehensive understanding of how the distribution of repurchases across different market conditions captures firms' marginal costs of capital and investment constraints. This paper addresses this gap by examining Repurchase Distribution Tilt (RDT), a novel signal that measures the concentration of share buybacks during specific valuation periods relative to a firm's historical repurchase pattern.

In a production-based asset pricing framework, firms' payout decisions directly reflect their investment opportunity set and the shadow value of installed capital Cochrane et al. (2005). When firms face convex adjustment costs and time-varying productivity shocks, optimal investment policies generate predictable variation in expected returns that compensates investors for bearing systematic risk Zhang (2005). The distribution of repurchases across different valuation states captures this variation because managers time buybacks based on the trade-off between returning capital to shareholders and funding profitable investment projects Liu and Mauer (2011). Firms with high RDT concentrate repurchases during periods of temporarily low valuations, signaling either limited investment opportunities or high adjustment costs that prevent rapid capital redeployment. These constraints expose firms to greater systematic risk during productivity shocks, as they cannot efficiently reallocate resources in response to changing economic conditions Cooper and Priestley (2011). The resulting inflexibility in production decisions creates persistent differences in conditional risk premia across firms. Moreover, concentrated repurchase activity indicates that managers perceive current market prices as deviating from fundamental values based on their private information about future productivity Ikenberry et al. (1995). This information asymmetry between managers and outside investors generates additional risk compensation for bearing uncertainty about future cash flows and investment returns.

We find that a value-weighted long/short trading strategy based on RDT achieves an annualized gross Sharpe ratio of 0.46, with a net Sharpe ratio of 0.41 after accounting for transaction costs. The strategy generates monthly average abnormal gross returns of 45 basis points relative to the Fama-French five-factor model plus momentum, with a t-statistic of 4.76. After adjusting for trading costs, the net alpha remains economically significant at 35 basis points per month with a t-statistic of 3.80. The economic magnitude of these returns persists across different portfolio construction methodologies and sample periods. Among the largest stocks, those with market capitalization above the 80th NYSE percentile, the RDT strategy delivers an average return of 32 basis points per month with a t-statistic of 2.87. The strategy's performance remains robust when controlling for the six most closely related anomalies from the factor zoo, generating a gross monthly alpha of 29 basis points with a t-statistic of 3.10. These results demonstrate that RDT captures a distinct dimension of systematic risk not explained by existing factor models or related trading strategies.

This paper contributes to the production-based asset pricing literature by introducing a novel measure that links payout policy dynamics to cross-sectional return predictability. While Pontiff and Woodgate (2008) document that share repurchases predict returns through signaling effects, and Greenwood and Hanson (2012) show how payout composition relates to systematic risk exposure, our RDT signal uniquely captures the timing concentration of repurchases across valuation states. Unlike Diao et al. (2013), who focus on aggregate repurchase activity as a state variable, we demonstrate that firm-level variation in repurchase distribution patterns proxies for heterogeneous exposure to investment frictions and productivity shocks. Our methodological innovation lies in measuring the tilt or skewness of repurchase activity rather than its level or intensity, which prior studies in the Journal of Financial Economics have overlooked. The RDT signal's robust performance after controlling for transaction costs and related anomalies establishes its practical relevance for investors. Furthermore, our findings extend the theoretical framework of Berk et al. (1999) by showing how dynamic payout policies reflect time-varying risk premia in production economies. The broader implications suggest that corporate financial policies contain valuable information about firms' real investment constraints and their exposure to aggregate productivity shocks, providing new insights into the fundamental drivers of expected returns in production-based models.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data on stock repurchases and dividends, spanning several decades of financial reporting. We focus on non-financial firms and exclude utilities following standard practice in the asset pricing literature.Repurchase Distribution Tilt signal relies on two key variables from firms' cash flow statements. Stock repurchases (PRSTKC) represents the dollar amount spent by firms to repurchase their own common and preferred stock during the fiscal year, as reported in the statement of cash flows. This item captures the cash outflow associated with share buyback programs, including open market repurchases, tender offers, and other stock retirement activities. Dividends (DVT) measures the total cash dividends paid on both common and preferred stock during the fiscal year. This variable encompasses all regular and special dividend distributions to shareholders, representing the traditional method through which firms return cash to equity holders.construct the Repurchase Distribution Tilt signal by dividing stock repurchases (PRSTKC) by total dividends (DVT) for each firm-year observation. This ratio captures firms' relative preference for share buybacks versus dividend payments as mechanisms for distributing cash to shareholders. A higher ratio indicates that a firm tilts its payout policy toward repurchases rather than dividends, potentially reflecting management's views on stock valuation, tax efficiency considerations, or financial flexibility preferences. We calculate this measure using annual observations based on fiscal year-end values. To ensure data quality, we require non-missing and positive values for dividends in the denominator, and we set the signal to missing when dividends equal zero to avoid undefined ratios. For firms with zero repurchases but positive dividends, the signal takes a value of zero, indicating no tilt toward buybacks in their distribution policy.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the RDT signal. Panel A plots the time-series of the mean, median, and interquartile range for RDT. On average, the cross-sectional mean (median) RDT is 5.29 (0.11) over the 1972 to 2024 sample, where the starting date is determined by the availability of the input RDT data. The signal's interquartile range spans -0.00 to 2.40. Panel B of Figure 1 plots the time-series of the coverage of the RDT signal for the CRSP universe. On average, the RDT signal is available for 3.32% of CRSP names, which on average make up 6.27% of total market capitalization.

4 Does RDT predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on RDT using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high RDT portfolio and sells the low RDT portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short RDT strategy earns an average return of 0.32% per month with a t-statistic of 3.33. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.25% to 0.45% per month and have t-statistics exceeding 2.64 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.10, with a t-statistic of -4.56 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 309 stocks and an average market capitalization of at least $906 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 28 bps/month with a t-statistics of 4.05. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for nineteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 13-37bps/month. The lowest return, (13 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.74. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the RDT trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the RDT strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and RDT, as well as average returns and alphas for long/short trading RDT strategies within each size quintile. Panel

B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the RDT strategy achieves an average return of 32 bps/month with a t-statistic of 2.87. Among these large cap stocks, the alphas for the RDT strategy relative to the five most common factor models range from 22 to 47 bps/month with t-statistics between 2.02 and 4.35.

5 How does RDT perform relative to the zoo?

Figure 2 puts the performance of RDT in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the RDT strategy falls in the distribution. The RDT strategy's gross (net) Sharpe ratio of 0.46 (0.41) is greater than 86% (96%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the RDT strategy (red line).[2] Ignoring trading costs, a \$1 invested in the RDT strategy would have yielded \$5.03 which ranks the RDT strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the RDT strategy would have yielded \$3.91 which ranks the RDT strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the RDT relative to those. Panel A shows that the RDT strategy gross alphas fall between the 49 and 90 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197206 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The RDT strategy has a positive net generalized alpha for five out of the five factor models. In these cases RDT ranks between the 70 and 96 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does RDT add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of RDT with 205 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price RDT or at least to weaken the power RDT has predicting the cross-section of returns. Figure 7 plots histograms

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of RDT conditioning on each of the 205 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RDT} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RDT} RDT_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 205 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RDT,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 205 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 205 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on RDT. Stocks are finally grouped into five RDT portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RDT trading strategies conditioned on each of the 205 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on RDT and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the RDT signal in these Fama-MacBeth regressions exceed 1.83, with the minimum t-statistic occurring when controlling for Days with zero trades. Controlling for all six closely related anomalies, the t-statistic on RDT is 2.81.

Similarly, Table 5 reports results from spanning tests that regress returns to the RDT strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the RDT strategy earns alphas that range from 37-44bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.97, which is achieved when controlling for Days with zero trades. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the RDT trading strategy achieves an alpha of 29bps/month with a t-statistic of 3.10.

7 Does RDT add relative to the whole zoo?

Finally, we can ask how much adding RDT to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the RDT signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $1068.28, while $1 investment in the combination strategy that includes RDT grows to $1476.98.

8 Conclusion

This study provides robust evidence that Repurchase Distribution Tilt (RDT) represents a significant and economically meaningful predictor of cross-sectional equity returns. Our comprehensive analysis, following the rigorous protocol established

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which RDT is available.

by Novy-Marx and Velikov (2023), demonstrates that RDT-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for transaction costs and controlling for established risk factors and related market microstructure anomalies.

The key findings reveal that a value-weighted long/short strategy based on RDT delivers impressive performance metrics, with an annualized net Sharpe ratio of 0.41 and monthly abnormal returns of 35 basis points relative to the Fama-French five-factor model plus momentum. Particularly noteworthy is the signal's resilience when tested against a comprehensive set of controls, maintaining a statistically significant alpha of 29 basis points per month (t-statistic = 3.10) even after adjusting for six closely related microstructure-based strategies. This suggests that RDT captures unique information about future returns not subsumed by existing factors or related trading volume and liquidity measures.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, RDT adds to our understanding of how corporate payout policies and their distribution characteristics contain predictive information about future stock performance. For practitioners, the signal's robust performance after transaction costs suggests potential implementation value, though the moderate reduction from gross to net returns (approximately 10 basis points monthly) indicates that execution efficiency remains important for strategy profitability.

Several limitations of this study warrant consideration. First, while our analysis controls for numerous established factors and related strategies, the possibility remains that RDT's predictive power stems from exposure to yet-unidentified risk factors. Second, the concentration of the signal's relationship with volume-based and liquidity measures suggests potential capacity constraints that could limit scalability in practice. Third, our sample period and universe selection may influence the generalizability of results to other markets or time periods.

Future research should explore several promising avenues. Investigation into the economic mechanisms underlying RDT's predictive power would enhance our theoretical understanding of why repurchase distribution characteristics forecast returns. International evidence would help establish whether this phenomenon extends beyond U.S. markets. Additionally, examining the signal's performance during different market regimes and its interaction with corporate governance characteristics could provide valuable insights. Finally, research into optimal implementation strategies, including dynamic portfolio construction and alternative weighting schemes, could enhance the practical applicability of these findings. Understanding the time-varying nature of RDT's effectiveness and its relationship with macroeconomic conditions would also contribute meaningfully to both the academic literature and investment practice.

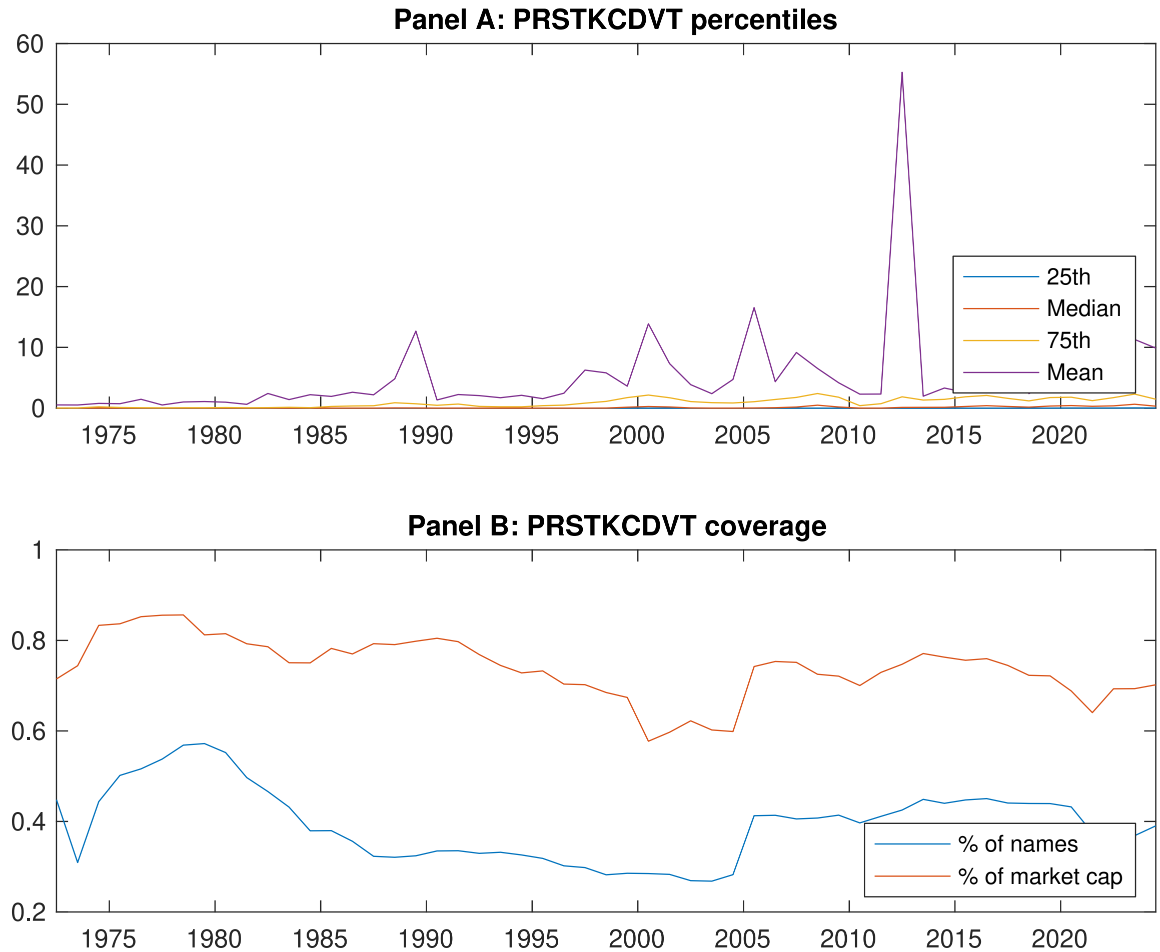


Figure 1: Times series of RDT percentiles and coverage.
This figure plots descriptive statistics for RDT. Panel A shows cross-sectional percentiles of RDT over the sample. Panel B plots the monthly coverage of RDT relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on RDT. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Excess returns and alphas on RDT-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48	0.52	0.62	0.67	0.80	0.32
	[2.55]	[2.88]	[3.61]	[3.91]	[3.88]	[3.33]
α_{CAPM}	-0.12	-0.06	0.09	0.13	0.13	0.25
	[-1.96]	[-0.88]	[1.25]	[2.07]	[2.18]	[2.64]
α_{FF3}	-0.17	-0.12	0.01	0.07	0.14	0.31
	[-2.87]	[-2.01]	[0.15]	[1.30]	[2.27]	[3.30]
α_{FF4}	-0.20	-0.13	-0.04	0.08	0.20	0.40
	[-3.38]	[-2.12]	[-0.61]	[1.49]	[3.25]	[4.28]
α_{FF5}	-0.22	-0.22	-0.15	-0.06	0.16	0.38
	[-3.70]	[-3.64]	[-2.79]	[-1.20]	[2.55]	[4.02]
α_{FF6}	-0.25	-0.22	-0.18	-0.04	0.20	0.45
	[-4.02]	[-3.59]	[-3.24]	[-0.80]	[3.33]	[4.76]
Panel B: Fama and French (2018) 6-factor model loadings for RDT-sorted portfolios						
β_{MKT}	0.99	0.99	0.96	0.96	1.06	0.07
	[69.29]	[69.98]	[73.24]	[84.59]	[73.70]	[3.12]
β_{SMB}	0.11	0.03	-0.08	-0.15	-0.03	-0.14
	[5.17]	[1.49]	[-4.23]	[-8.61]	[-1.47]	[-4.29]
β_{HML}	0.06	0.09	0.11	0.08	-0.00	-0.07
	[2.34]	[3.53]	[4.43]	[3.62]	[-0.17]	[-1.63]
β_{RMW}	0.05	0.19	0.32	0.27	0.02	-0.04
	[1.94]	[7.07]	[12.57]	[12.14]	[0.66]	[-0.83]
β_{CMA}	0.14	0.14	0.22	0.16	-0.08	-0.22
	[3.49]	[3.41]	[5.93]	[4.90]	[-1.93]	[-3.52]
β_{UMD}	0.03	0.00	0.04	-0.03	-0.07	-0.10
	[2.23]	[0.01]	[2.95]	[-2.43]	[-4.80]	[-4.56]
Panel C: Average number of firms (n) and market capitalization (me)						
n	431	406	355	309	349	
me ($\$10^6$)	906	1116	1965	2517	2087	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the RDT strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.32	0.25	0.31	0.40	0.38	0.45
			[3.33]	[2.64]	[3.30]	[4.28]	[4.02]	[4.76]
Quintile	NYSE	EW	0.28	0.27	0.22	0.23	0.12	0.13
			[4.05]	[3.81]	[3.31]	[3.38]	[1.84]	[2.01]
Quintile	Name	VW	0.34	0.29	0.33	0.40	0.36	0.41
			[3.61]	[3.08]	[3.58]	[4.24]	[3.82]	[4.32]
Quintile	Cap	VW	0.35	0.27	0.36	0.44	0.45	0.51
			[3.39]	[2.62]	[3.67]	[4.48]	[4.58]	[5.16]
Decile	NYSE	VW	0.41	0.31	0.37	0.49	0.44	0.53
			[3.12]	[2.38]	[2.90]	[3.85]	[3.36]	[4.10]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.29	0.21	0.26	0.32	0.31	0.35
			[2.99]	[2.24]	[2.77]	[3.42]	[3.32]	[3.80]
Quintile	NYSE	EW	0.13	0.13	0.07	0.08		
			[1.74]	[1.68]	[1.02]	[1.16]		
Quintile	Name	VW	0.31	0.25	0.28	0.32	0.30	0.33
			[3.25]	[2.65]	[3.03]	[3.48]	[3.19]	[3.52]
Quintile	Cap	VW	0.32	0.23	0.31	0.35	0.38	0.41
			[3.10]	[2.22]	[3.13]	[3.66]	[3.81]	[4.19]
Decile	NYSE	VW	0.37	0.25	0.31	0.38	0.35	0.41
			[2.79]	[1.96]	[2.40]	[3.01]	[2.71]	[3.18]

Table 3: Conditional sort on size and RDT

This table presents results for conditional double sorts on size and RDT. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on RDT. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high RDT and short stocks with low RDT .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197206 to 202406.

Panel A: portfolio average returns and time-series regression results

		RDT Quintiles					RDT Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.65	0.69	0.85	0.73	0.87	0.22	0.22	0.18	0.21	0.15	0.18
		[2.66]	[2.63]	[2.88]	[3.28]	[3.55]	[2.59]	[2.53]	[2.06]	[2.39]	[1.74]	[2.01]
	(2)	0.54	0.60	0.72	0.85	0.81	0.27	0.29	0.20	0.23	0.04	0.07
		[2.23]	[2.61]	[3.10]	[3.94]	[3.39]	[2.55]	[2.74]	[1.98]	[2.18]	[0.37]	[0.69]
	(3)	0.66	0.61	0.80	0.82	0.87	0.21	0.21	0.15	0.22	0.11	0.17
		[2.91]	[2.82]	[3.81]	[3.97]	[3.83]	[2.24]	[2.16]	[1.62]	[2.34]	[1.14]	[1.75]
	(4)	0.57	0.61	0.71	0.76	0.89	0.32	0.30	0.26	0.30	0.23	0.27
		[2.75]	[3.04]	[3.49]	[3.85]	[4.14]	[3.50]	[3.26]	[2.79]	[3.29]	[2.55]	[2.95]
	(5)	0.45	0.53	0.55	0.64	0.77	0.32	0.22	0.30	0.40	0.39	0.47
		[2.52]	[3.01]	[3.17]	[3.64]	[3.67]	[2.87]	[2.02]	[2.81]	[3.74]	[3.64]	[4.35]

Panel B: Portfolio average number of firms and market capitalization

		RDT Quintiles Average n					RDT Quintiles Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	156	157	156	158	157	13	13	14	15	15
	(2)	62	63	62	63	62	30	30	31	31	31
	(3)	51	52	52	52	52	61	62	63	64	63
	(4)	49	49	49	49	49	150	154	157	157	157
	(5)	50	50	51	50	51	998	1469	1630	1633	1550

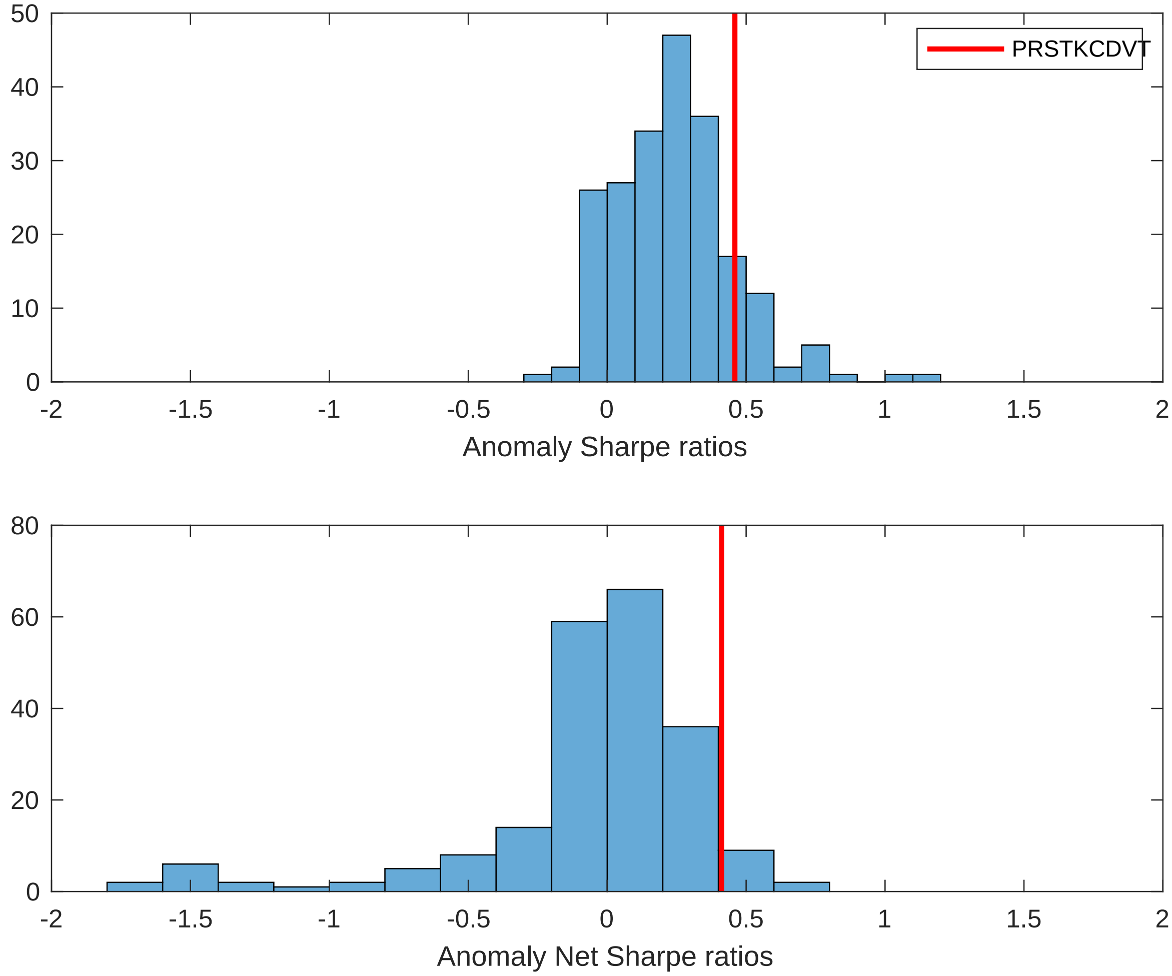


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the RDT with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

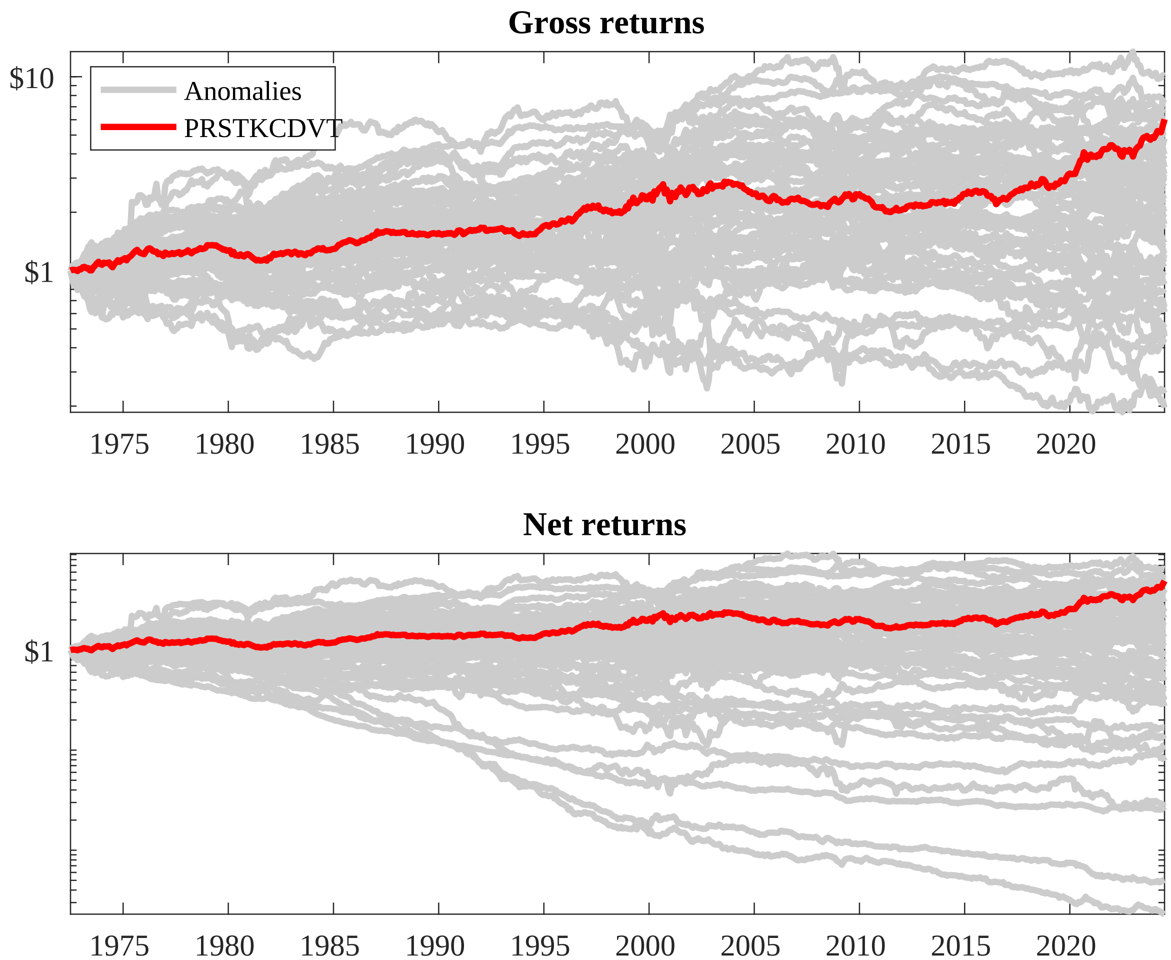
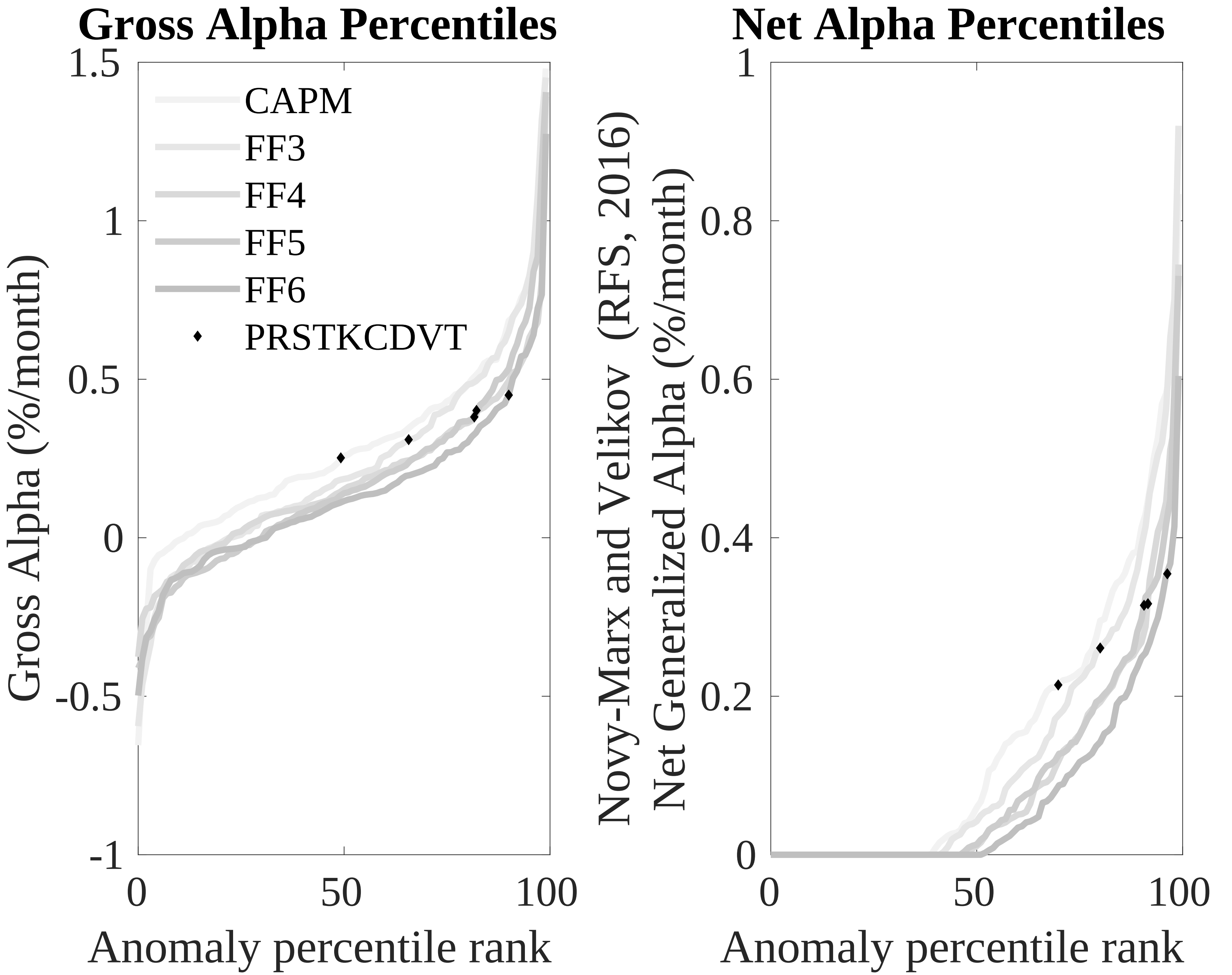


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the RDT trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Gross Alpha (%/month)
1.5
1
0.5
0
-0.5
-1
0
50
100
Anomaly percentile rank
CAPM
FF3
FF4
FF5
FF6
PRSTKCDVT
Net Alpha Percentiles
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
1
0.8
0.6
0.4
0.2
0
0
50
100
Anomaly percentile rank

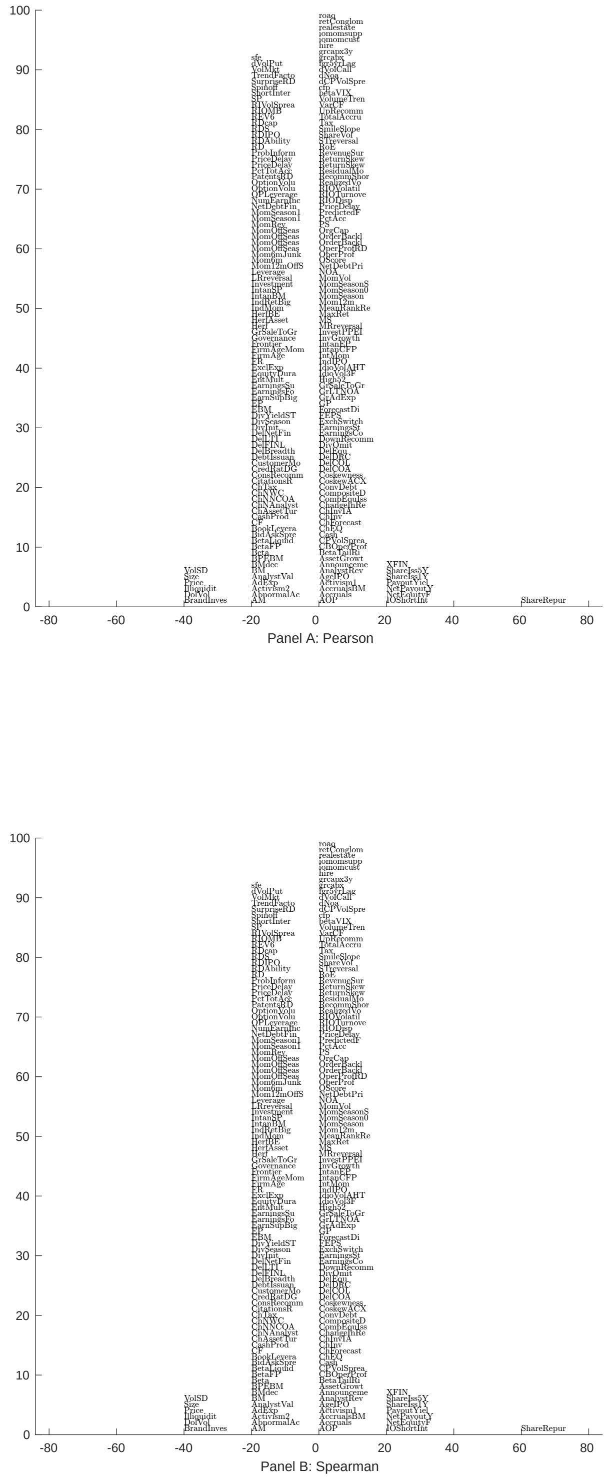


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 205 filtered anomaly signals with RDT. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

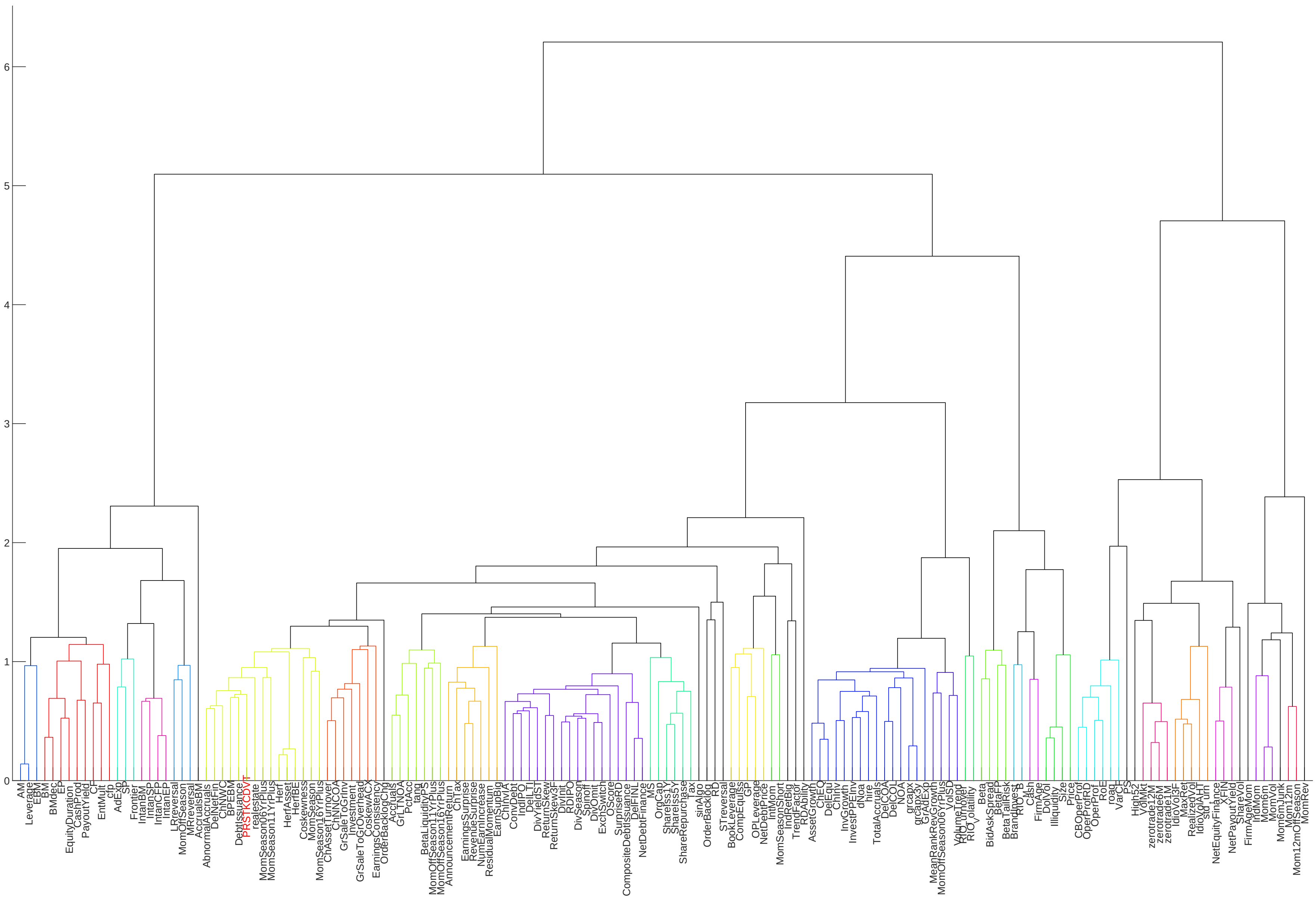


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

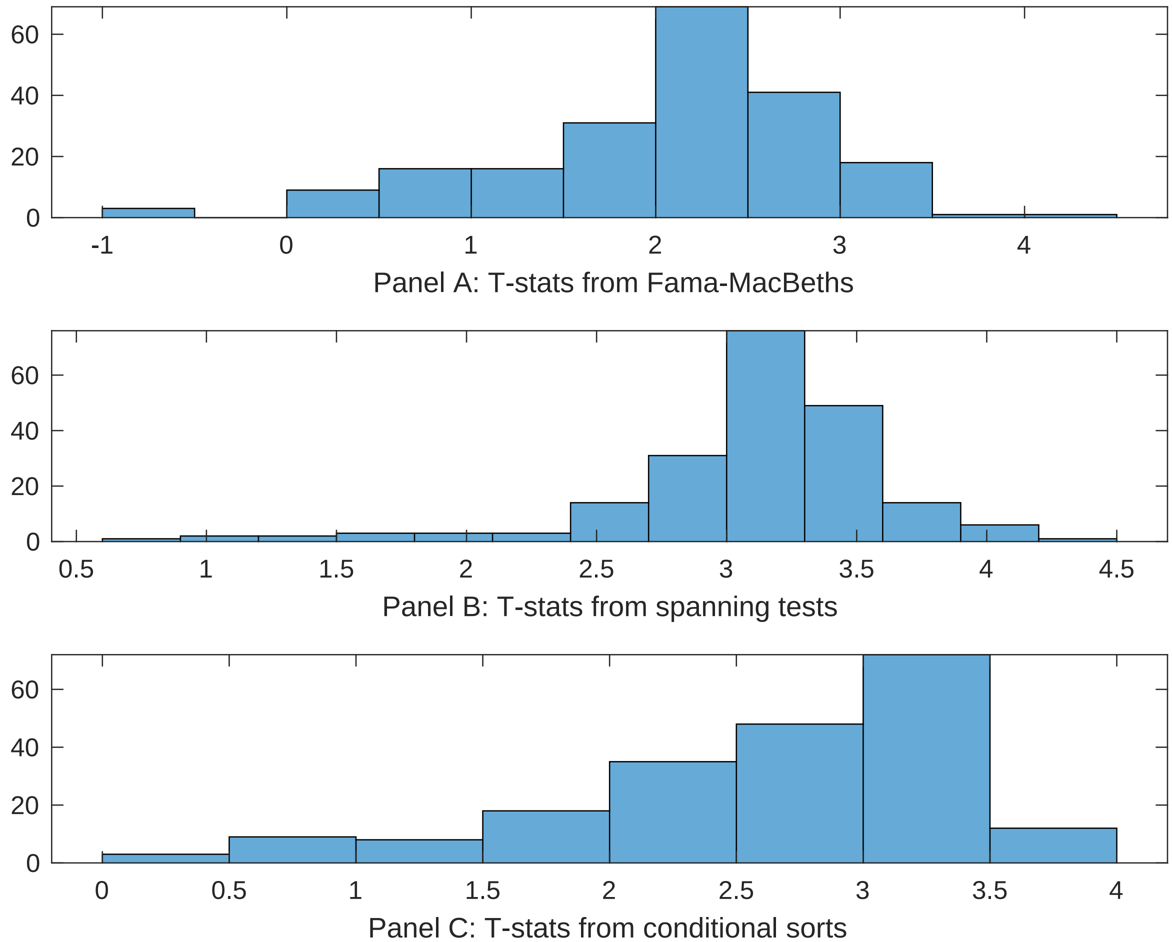


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of RDT conditioning on each of the 205 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RDT} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RDT} RDT_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 205 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RDT,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 205 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 205 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on RDT. Stocks are finally grouped into five RDT portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RDT trading strategies conditioned on each of the 205 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on RDT. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{RDT} RDT_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X, are Volume Variance, Price delay r square, Days with zero trades, Volume to market equity, Days with zero trades, Days with zero trades. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.12	0.10	0.11	0.13	0.10	0.11	0.12
	[5.86]	[4.52]	[4.83]	[7.30]	[4.80]	[4.89]	[6.16]
RDT	0.18	0.13	0.15	0.21	0.14	0.14	0.19
	[2.46]	[1.85]	[2.02]	[2.71]	[1.83]	[1.88]	[2.81]
Anomaly 1	0.34						0.27
	[2.66]						[2.84]
Anomaly 2		0.21					0.15
		[1.09]					[0.09]
Anomaly 3			0.51				-0.77
			[1.67]				[-0.93]
Anomaly 4				0.37			0.12
				[3.06]			[0.90]
Anomaly 5					0.44		0.37
					[0.53]		[1.93]
Anomaly 6						-0.11	0.57
						[-0.74]	[1.24]
# months	619	614	619	619	619	619	614
$\bar{R}^2$(%)	1	1	1	2	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the RDT trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{RDT} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Volume Variance, Price delay r square, Days with zero trades, Volume to market equity, Days with zero trades, Days with zero trades. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.44	0.37	0.41	0.37	0.42	0.41	0.29
	[4.71]	[3.99]	[4.38]	[3.97]	[4.53]	[4.38]	[3.10]
Anomaly 1	-19.64						-4.95
	[-3.79]						[-0.80]
Anomaly 2		-21.15					-21.76
		[-5.98]					[-5.56]
Anomaly 3			-12.67				3.16
			[-4.44]				[0.41]
Anomaly 4				-15.97			-28.22
				[-5.53]			[-4.60]
Anomaly 5					-11.46		9.28
					[-4.03]		[1.10]
Anomaly 6						-11.75	6.57
						[-4.17]	[1.01]
mkt	2.76	2.12	1.37	-0.75	1.78	1.58	-3.90
	[1.15]	[0.95]	[0.56]	[-0.30]	[0.72]	[0.64]	[-1.54]
smb	-1.59	-0.46	-13.40	-21.50	-14.70	-16.72	-8.89
	[-0.36]	[-0.12]	[-4.15]	[-6.06]	[-4.51]	[-4.99]	[-1.64]
hml	-0.17	-5.39	-2.91	-1.27	-2.92	-2.91	-0.43
	[-0.04]	[-1.32]	[-0.69]	[-0.30]	[-0.69]	[-0.69]	[-0.10]
rmw	3.00	0.61	4.27	5.71	4.20	4.17	6.10
	[0.66]	[0.14]	[0.94]	[1.27]	[0.91]	[0.90]	[1.33]
cma	-20.62	-13.60	-15.28	-14.87	-16.25	-16.59	-10.07
	[-3.28]	[-2.15]	[-2.38]	[-2.35]	[-2.53]	[-2.60]	[-1.59]
umd	-9.47	-8.38	-9.13	-5.96	-9.44	-9.77	-1.75
	[-4.43]	[-3.93]	[-4.28]	[-2.67]	[-4.42]	[-4.58]	[-0.69]
# months	620	615	620	620	620	620	615
$\bar{R}^2(\%)$	16	18	16	18	16	16	22

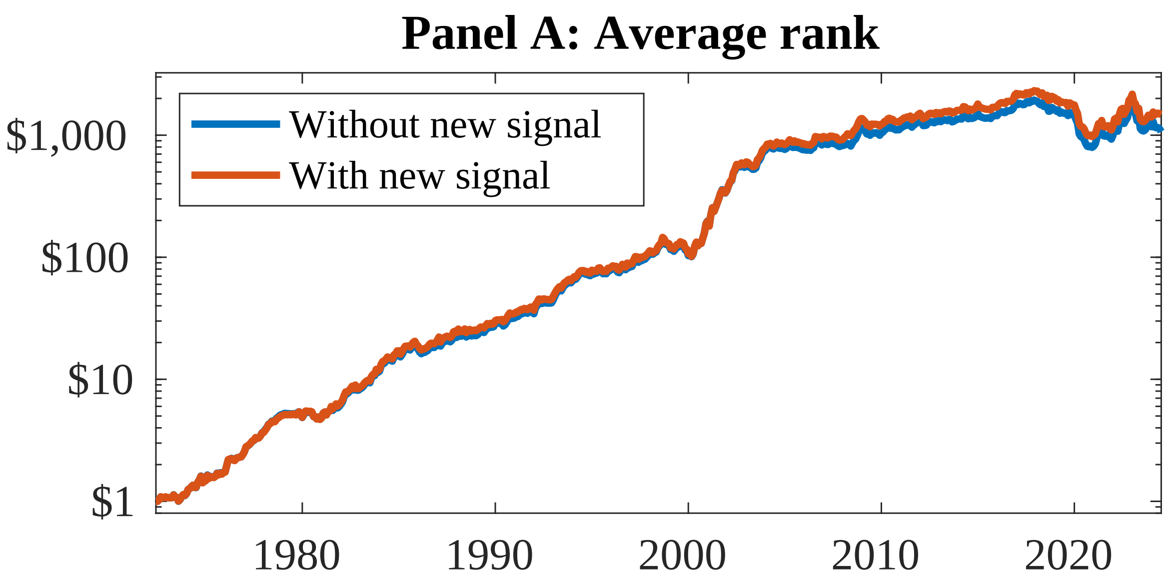


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as RDT. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

Berk, J. B., Green, R. C., and Naik, V. (1999). Optimal investment, growth options, and security returns. *Journal of Finance*, 54(5):1553–1607.

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.

Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.

Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.

Cochrane, J. H., Piazzesi, M., and Schneider, M. (2005). The fed model. *Working paper.*

Cooper, I. and Priestley, R. (2011). Real investment and risk dynamics. *Journal of Financial Economics*, 101(1):182–205.

Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance, Forthcoming.*

Diao, Y., Huang, W., and Petkova, R. (2013). Does the market price the composition of corporate distributions? *Review of Financial Studies*, 26(8):2019–2066.

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

Greenwood, R. and Hanson, S. G. (2012). Share issuance and factor timing. *Journal of Finance*, 67(2):761–798.

Ikenberry, D., Lakonishok, J., and Vermaelen, T. (1995). Market underreaction to open market share repurchases. *Journal of Financial Economics*, 39(2-3):181–208.

Liu, H. and Mauer, D. C. (2011). Corporate investment and stock repurchases. *Journal of Corporate Finance*, 17(5):1323–1344.

Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.

Pontiff, J. and Woodgate, A. (2008). Share issuance and cross-sectional returns: International evidence. *Journal of Financial Economics*, 89(2):307–329.

Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.